FINN_SC illustrated tutorial

Biomass-burning emissions of Santa Catarina, Brazil, from the Fire INventory from NCAR: the science, the code and every step to reproduce the results

for scientists | for students | for decision makers | Python · R · Colab | version of 11 Nov 2026

Repository: https://github.com/jessicajcss/FINN_SC (a Zenodo DOI will replace this address). Companion of the article submitted to Environmental Science & Technology, of the full-length article v7.2 and of the 2025 state inventory report. Code and text: MIT licence. Authors: J. C. dos Santos Silva and co-authors (see CITATION.cff).

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1 How to use this tutorial

Chapter 1 / 14 · reading time: 3 min · everyone

IN 30 SECONDS

Read chapters 2 to 4 to understand the study, chapters 5 to 7 to run it, and the rest when you need it. Every chapter opens with this orange box, so you can stop after it and still keep the main point.

After this chapter you will be able to:

- [ ] choose your reading path
- [ ] recognise the five kinds of boxes
- [ ] find any number of the article in the code

Three reading paths

If you want to…	Read
know what the study found, in plain words	2
understand how satellites and FINN estimate emissions	3 and 4
run everything and check the results	5, 6 and 7
trace one published number to its code	8
understand the statistics	9
compare FINN versions and other inventories	10
build the inventory of another region	11 and 12

What each box means

KEY IDEA — The one idea to keep from the section.

WATCH OUT — A common mistake or a limit of the data.

DO IT — Commands to type, in order.

CHECK — How to know that it worked.

DEEP DIVE (OPTIONAL) — Extra detail for specialists; safe to skip.

KEY IDEA

Nothing in this tutorial is new: every number comes from a file in `results/`, written by a script of the repository and checked by an automated test. When the text gives a number, the chapter or the table of chapter 8 tells you where it comes from.

CHECK YOURSELF

Which box can you skip without losing the main point?

The grey “deep dive” box.

2 The study on one page

Chapter 2 / 14 · reading time: 5 min · everyone

IN 30 SECONDS

Fire emissions in Santa Catarina vary a lot between years, but **three methodological choices change the totals as much as, or more than, the years do**: adding VIIRS detections, the FINN version and the forest class. Comparisons between years and municipalities hold; absolute totals carry these uncertainties.

After this chapter you will be able to:

- [ ] state the four main findings
- [ ] explain why absolute totals need care
- [ ] read Figure 4 of the article

3,549–8,296
fire events per year, 2012–2025, with no monotonic trend

3.55 Mt
CO_2 in 2025 (FINN v2.5.1, MODIS+VIIRS)

2.9×
ratio between the highest (2020) and lowest (2015) annual CO_2

53.6%
of grassland-dominated 2 km cells burned in 7 or more of 14 years

Four findings

1. **Year-to-year variability.** Events ranged from 3,549 (2015) to 8,296 (2016) and were fewer in wetter fire seasons (Spearman $\rho = -0.68$ with fire-season precipitation).
2. **Recurrence on the highland grasslands.** Fire returns to the same places: 53.6% of the 2 km cells dominated by grassland burned in at least 7 of 14 years, mostly on the Serrana plateau.
3. **Methodological choices rival the years.** Adding VIIRS raised CO_2 by 40% to 56%. For the same 2020 MODIS detections, FINN v1.5 gave 2.8 times the FINN v2.5 CO_2. A tropical forest class would multiply the FINN v2.5 CO_2 by 2.4 to 2.7. FINN burned area exceeded the mapped burned area by median factors of 17 (MapBiomas Fire) and 34 (MODIS MCD64A1).
4. **Earlier estimates count assumed fires.** FINN v1.5 adds a half-size copy of each fire on the next day; the earlier state inventory, BRAIN and the FINN v1.5 totals of Teixeira et al. (2025) all include these copies, 27.0% to 32.4% of the yearly CO_2 in 2002–2020.

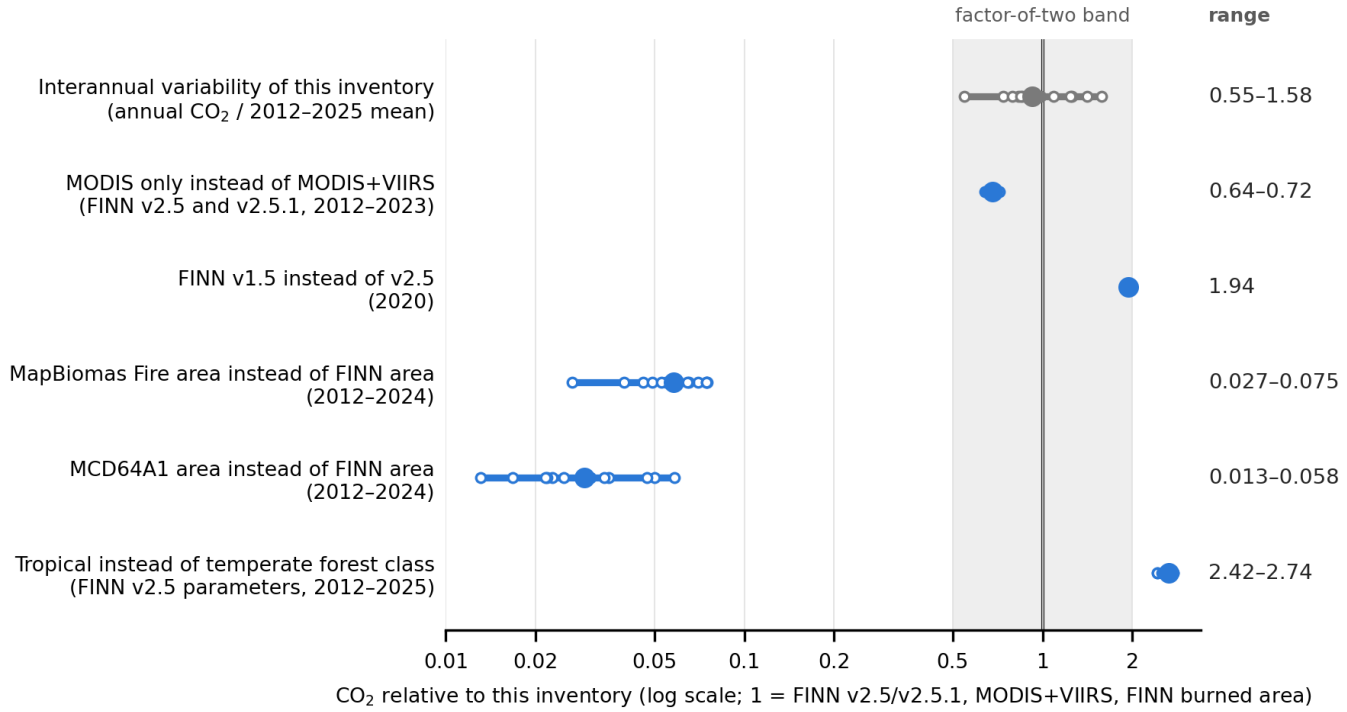


Figure 4 of the ES&T manuscript. Grey: annual CO_2 relative to the 2012–2025 mean. Coloured: CO_2 with each alternative choice relative to the inventory. Script `python/analysis/fig_v71.py`; values in `results/tables/sensitivity_factors.csv` (`a15_v7_review.py`) and `forest_class_scenario_annual.csv` (`a19_v72_followup.py`).

⚠ WATCH OUT

"Fire event" here is a FINN daily burned-area polygon, a model unit. It is neither a field-verified fire nor an INPE hotspot, so counts from other sources are compared only with the definitions side by side (chapter 3).

CHECK YOURSELF

Which choice changed the 2020 CO_2 the most?

The FINN version: FINN v1.5 gave 2.8 times the FINN v2.5 MODIS value.

Do the rankings of municipalities depend on these choices?

Much less than the totals: relative patterns were robust.

3 Fire, satellites and emissions

Chapter 3 / 14 · reading time: 10 min · students and non-specialists

IN 30 SECONDS

Satellites detect the heat of active fires. FINN turns each detection into burned area, multiplies it by fuel, burned fraction and emission factors, and writes one row (a **record**) per detection and vegetation type. We count **fire events**, the daily burned-area polygons, after removing persistent heat sources.

After this chapter you will be able to:

- [ ] explain the FINN equation
- [ ] tell a record from a detection and an event
- [ ] convert FINN units to tonnes

3.1 Why estimate fire emissions

Open burning releases CO_2, CO, fine particles ($PM_{2.5}$), nitrogen oxides and volatile organic compounds. Air-quality models, health studies and climate inventories need these emissions by place and day, and ground measurements cannot cover a whole State. Satellite inventories fill this gap.

3.2 How satellites see fire

MODIS (on the Terra and Aqua satellites) detects active fires in 1 km pixels; **VIIRS** (Suomi NPP, NOAA-20) does so in 375 m pixels and sees smaller fires (Giglio et al., 2016; Schroeder et al., 2014). Each satellite passes over a place about twice a day, so fires that start and end between passes, or under clouds, are missed.

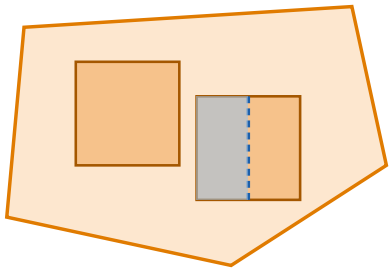
3.3 The FINN equation



per record, per compound

For each detection, FINN assigns a burned area (A) from the pixel and the vegetation map, the fuel loading (B) and the fraction burned (CF) of the vegetation type, and an emission factor (EF) for each compound (Wiedinmyer et al., 2011, 2023). The vegetation types are six generic classes (GENVEG): grassland and savanna, woody savanna and shrubland, tropical forest, temperate forest, boreal forest and cropland.

3.4 Record, detection and event



record
one row of a FINN file: a detection, or the part of it in one vegetation type

detection
one satellite pixel on one day (day + FIREID)

fire event
the daily burned-area polygon (day + POLYID): the unit counted here

KEY IDEA

Counts change by unit. In 2025: 8,581 records, 5,489 detections and 5,160 fire events (5,121 after the screening below). This repository counts **fire events**, each assigned to the municipality that holds most of its area (`a01_events_screening.py` → `state_counts.csv`).

3.5 Persistent thermal anomalies

Some heat sources are not vegetation fires: industrial plants, mines, burning coal waste. A location with detections on 16 or more days of a calendar year within 1 km is treated as a persistent thermal anomaly, the criterion of the MODIS product (Giglio et al., 2021). These events are removed from counts and maps and kept in the emission totals, where their share is small (at most 2.5% of CO_2, in 2020).

3.6 Units: from moles per day to tonnes

FINN files give gases in mol/day and aerosols (OC, BC, $PM_{2.5}$) and a few mass fields in kg/day. Tonnes of a gas = mol × molar mass (g/mol) × 10^{-6}; tonnes of an aerosol = kg × 10^{-3}. In the FINN v2.5 files, `NOXasNO`, `NMOC` and `NMHC` are already mass fields (kg/day); NO_x is reported as the mass of NO and VOC as NMOC (`python/finn_sc/convert_units.py`, checked by `a09_emission_factor_check.py`).

WATCH OUT

FINN burned area is the area assigned to each detection by the model, much larger than the area mapped from burn scars: median 17 times MapBiomas Fire and 34 times MCD64A1 in Santa Catarina. Use it as a model term, not as mapped burned area.

CHECK YOURSELF

A detection split between forest and cropland gives how many records?

Two.

How many tonnes are 1,000 mol of CO_2?

1,000 × 44.01 × 10^{-6} = 0.044 t.

4 FINN versions and their conventions

Chapter 4 / 14 · reading time: 8 min · users of FINN data

IN 30 SECONDS

This inventory uses **FINN v2.5** (2012–2021) and **v2.5.1** (2022–2025) with MODIS and VIIRS. Two conventions of the model matter in Santa Catarina: FINN v1.5 adds next-day copies of fires (v2.5 does not, at these latitudes), and both versions classify the State's forests as **temperate** because it lies south of 23.5° S.

After this chapter you will be able to:

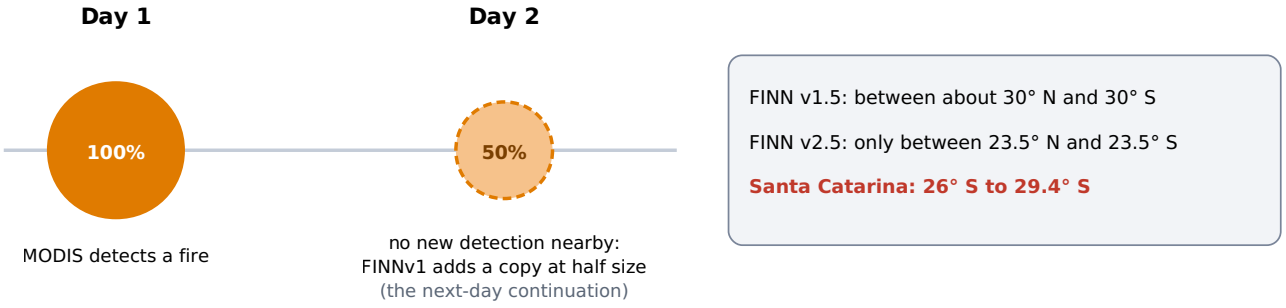
- [ ] pick the right FINN file for a year
- [ ] explain the next-day continuation
- [ ] explain the forest-class boundary
- [ ] follow the data terms of NSF NCAR

4.1 Which file for which year

Version	Detections	Years	File (GEOS-Chem, each fire)
v1.5	MODIS	2002–2020	`FINNv1.5_<year>.GEOSCHEM.tar.gz` (annual, every year)
v2.5	MODIS	2003–2021	`FINNv2.5_mod_GEOSCHEM_<year>_c2021…txt.gz`
v2.5	MODIS+VIIRS	2012–2021	`FINNv2.5_modvrs_GEOSCHEM_<year>_c2021…txt.gz`
v2.5.1	both	2022–2023	monthly files
v2.5.1 NRT	MODIS+VIIRS	2024–	daily near-real-time files

The exact addresses are built by `R/utils_finn.R` and `python/finn_sc/download.py`. The FINN v1.5 file of 2002 is also a `.tar.gz` archive; an older `.txt.gz` address returns “404 Not Found” (corrected on 11 Nov 2026).

4.2 The next-day continuation of FINN v1.5



Because MODIS does not observe low and middle latitudes every day, FINNv1 counts each detection for two days, the second at half its size, unless the place is detected again; its authors noted that this assumption had yet to be verified (Wiedinmyer et al., 2011). In 2020, 1,893 of the 4,609 FINN v1.5 records in Santa Catarina were such continuations, 27.0% of the CO_2 (`a17_v72_review.py`). FINN v2.5 keeps the rule only between 23.5° N and 23.5° S (Wiedinmyer et al., 2023), and our FINN v2.5 and

v2.5.1 records confirm it: 1 of 151,443 day-location pairs has the half-size signature (`a20_finn15_correction.py`).

4.3 The forest class depends on latitude

FINN v2.5 classes evergreen broadleaf and mixed forests as tropical forest only between 23.5° N and 23.5° S, and as temperate forest elsewhere (Wiedinmyer et al., 2023, Table 1). Santa Catarina (25.96° S to 29.36° S) falls outside, although its forests belong to the Atlantic Forest. The class sets the fuel loading: in South America, 26,755 g m^{-2} for tropical and 7,400 g m^{-2} for temperate forest. With the tropical class, each forest detection would emit 3.93 times more CO_2, and the State's annual CO_2 would be 2.42 to 2.74 times larger (`a19_v72_followup.py`), an upper bound because the class also holds deciduous forest.

4.4 Data terms

⚠ WATCH OUT

- FINN v1.5: the FINN page of NSF NCAR (https://www.acom.ucar.edu/Data/fire/) asks users to submit their name and contact information before downloading. Do this before running any FINN v1.5 download.
- FINN v2.5 and v2.5.1: NSF NCAR Research Data Archive, dataset ds312.9 (Wiedinmyer and Emmons, 2022, doi:10.5065/XNPA-AF09). UCAR's data terms grant use under the Creative Commons Attribution 4.0 licence unless otherwise stated and ask users to cite the data with the repository's recommended citation.
- Cite the model papers (Wiedinmyer et al., 2011, 2023) in any work that uses FINN.

CHECK YOURSELF

Does FINN v2.5 add next-day continuations in Santa Catarina?

No: only between 23.5° N and 23.5° S; the State is south of that band.

Why are the State's forests “temperate” in FINN?

Because FINN v2.5 uses the tropical class only within 23.5° of the equator.

5 Inside the repository

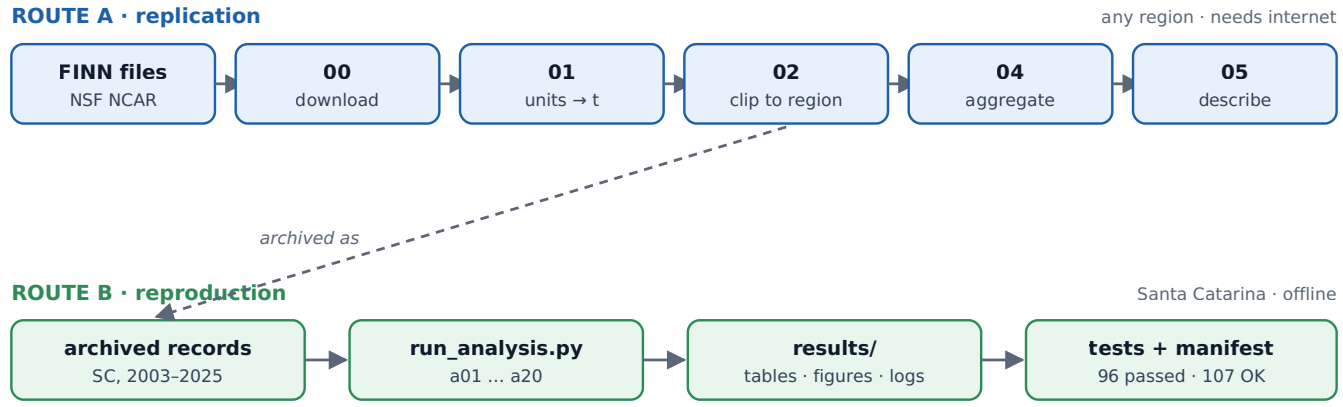
Chapter 5 / 14 · reading time: 5 min · anyone who will run the code

IN 30 SECONDS

There are two routes. **Route B** recomputes every number of the article offline from the archived records. **Route A** downloads FINN files from NSF NCAR and builds the records, for Santa Catarina or any other region.

After this chapter you will be able to:

- [ ] find any folder
- [ ] choose route A or route B




```
FINN_SC/
├── README.md, README.pt-BR.md      start here
├── TUTORIAL_FINN_SC_en.pdf, _pt.pdf  this tutorial
├── config/config.yml               every parameter (years, version, region)
├── data/derived/                   archived FINN records of SC (route B input)
├── data/external/                  MapBiomas, MCD64A1, ERA5-Land, IBGE, FINN v1.5 records
├── data/inputs/                    IBGE 2024 municipal mesh of SC
├── python/00_ … 06_*.py            route A in Python
├── python/analysis/a01 … a20       route B: one script per group of results
├── python/run_analysis.py          route B in one command
├── R/                              route A in R; kernel density
├── gee/                            Google Earth Engine recipes
├── notebooks/                      route A notebooks; BRAIN comparison
├── tools/                          maps, FINN v1.5 tools, manifest check
├── tests/                          96 automated tests
├── results/                        published tables, figures, logs
└── docs/                           methods, provenance, data dictionary
```

KEY IDEA

Route B starts from `data/derived/finn_sc_records_2003_2025.csv.gz`: 258,694 FINN v2.5 and v2.5.1 records inside Santa Catarina. It needs no internet and no FINN download.

What each analysis script answers

Script	Question
`a01`	How many records, detections and events? Which are persistent anomalies?
`a02`	Annual series, Table 2, effect of VIIRS, trend tests
`a03`	Municipalities and mesoregions in 2025; seasonality
`a04`	Area and emissions by vegetation type (GENVEG)
`a05`	FINN area vs MapBiomas Fire
`a06`	Fire activity vs fire-season climate
`a07`	Municipal land cover vs event density
`a08–a09`	Report indicators; unit-conversion check
`a10`	Figures of the earlier manuscript and report
`a11–a12`	Co-authors' questions; Earth Engine outputs (MCD64A1, ERA5-Land)
`a13`	Fire recurrence in 2 km cells and land cover
`a14`	Validity checks (FINN vegetation vs MapBiomas, bootstrap)
`a15`	Robustness: exact trend tests, VIIRS ratio, sensitivity factors
`a16–a17`	FINN v1.5 vs v2.5 in 2020; next-day continuations
`a18`	Exports of municipalities and mesoregions (Comex Stat)
`a19`	Forest-class scenario; NO_x of the earlier inventory
`a20`	FINN v1.5 inventories with and without the continuations, 2002–2020

CHECK YOURSELF

Which route needs internet?

Route A.

6 Hands on: reproduce the results

Chapter 6 / 14 · reading time: 10 min · anyone with Python

IN 30 SECONDS

Three commands: install, run, check. If the tests say **96 passed** and the manifest says **107 files: OK 107**, your computer reproduced every published number.

After this chapter you will be able to:

- [ ] install the environment
- [ ] run route B
- [ ] verify the outputs against the published ones

DO IT · 1. INSTALL (ONCE)

```
git clone https://github.com/jessicajcss/FINN_SC.git
cd FINN_SC
conda env create -f environment.yml && conda activate finn
# or: python -m pip install -r requirements.txt
Rscript install_r_packages.R   # optional: R for the kernel density
```

DO IT · 2. RUN ROUTE B

```
python python/run_analysis.py --no-r
# without --no-r it also runs R (the annual kernel sensitivity takes about 40 min)
```

Shortcuts: `make reproduce`, `./run_all.sh analysis` (Linux, macOS) or `run_all.bat analysis` (Windows).

CHECK · 3. VERIFY

```
python -m pytest -q tests          # 96 passed
python tools/verify_manifest.py    # 107 files: OK 107
```

The manifest (`results/MANIFEST.sha256`) holds the SHA-256 of every published table and log (figures are left out, because PNG files carry the library version in their metadata). OK means identical bytes; EQUIVALENT means that every number agrees within a relative tolerance of 10^{-9}, as happens when another library version prints a different last digit. Run the check in a fresh clone and commit nothing before it.

Where the outputs go

Folder	Content
`results/tables/`	CSV tables, one or more per script
`results/figures/`	PNG figures (English `_en`, Portuguese without suffix; `_nn` = no note inside the image)
`results/logs/`	one text log per script, with every number printed

⚠ WATCH OUT

Run the commands from the repository root (the folder that holds `README.md`). Most errors of “file not found” come from running them inside a subfolder.

CHECK YOURSELF

What does “107 files: OK 107” mean?

Every published output was reproduced byte for byte.

7 Reading the main figures

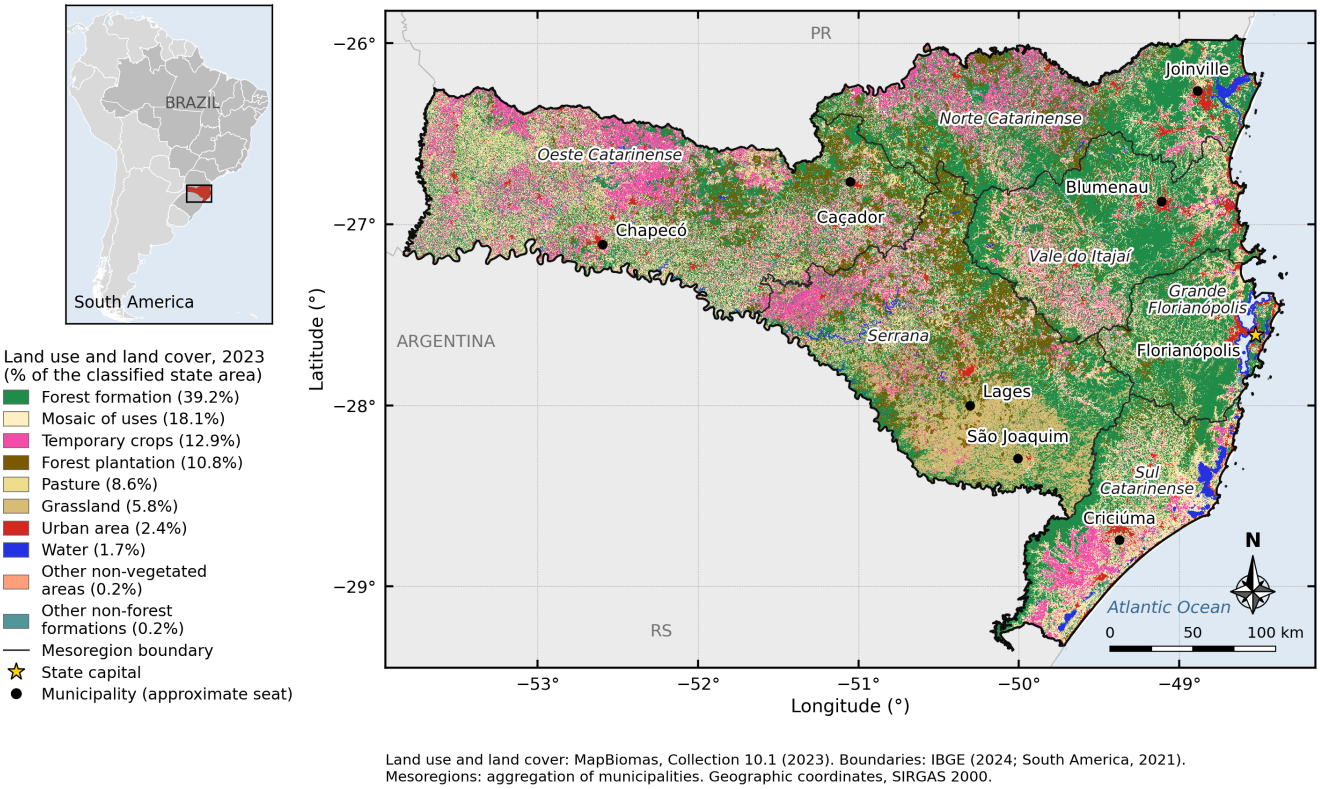
Chapter 7 / 14 · reading time: 8 min · everyone

IN 30 SECONDS

Four figures tell the story: where the State is (Figure 1), how the years changed (Figure 2), where fire recurs (Figure 3) and how much the methodological choices weigh (Figure 4, chapter 2).

After this chapter you will be able to:

- [ ] read each panel
- [ ] name the script behind each figure



Land use and land cover: MapBiomas, Collection 10.1 (2023). Boundaries: IBGE (2024; South America, 2021). Mesoregions: aggregation of municipalities. Geographic coordinates, SIRGAS 2000.

Figure 1. Santa Catarina, its six mesoregions and land use in 2023 (MapBiomas Collection 10.1). Script `tools/study_area_map.py`.

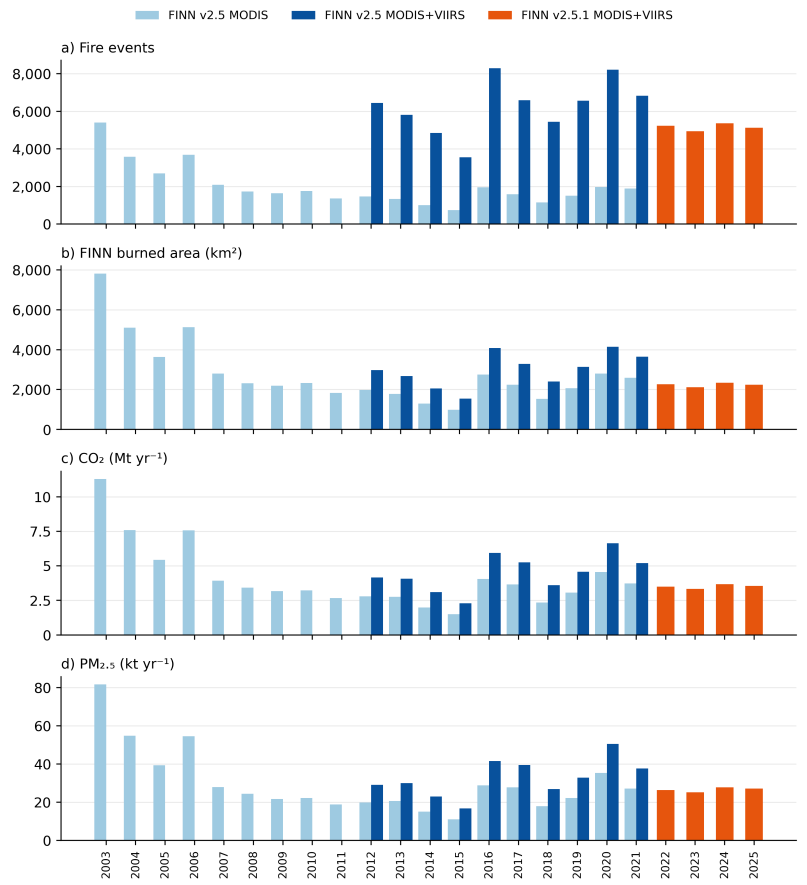


Figure 2. Annual events, FINN burned area, CO_2 and $PM_{2.5}$ by FINN configuration. Script `python/analysis/figures_est_en.py`; data `series_state_all_products.csv` (a02).

KEY IDEA

Compare years within one colour (one FINN configuration). Jumping between colours mixes the change of the fires with the change of the method.

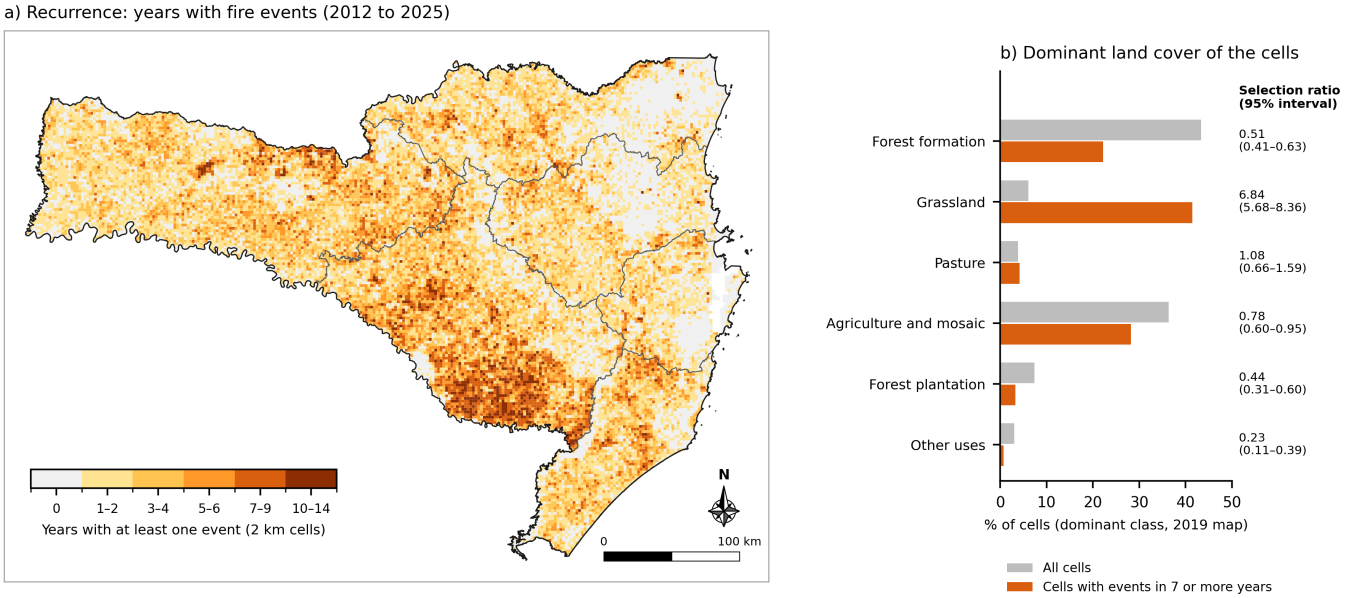
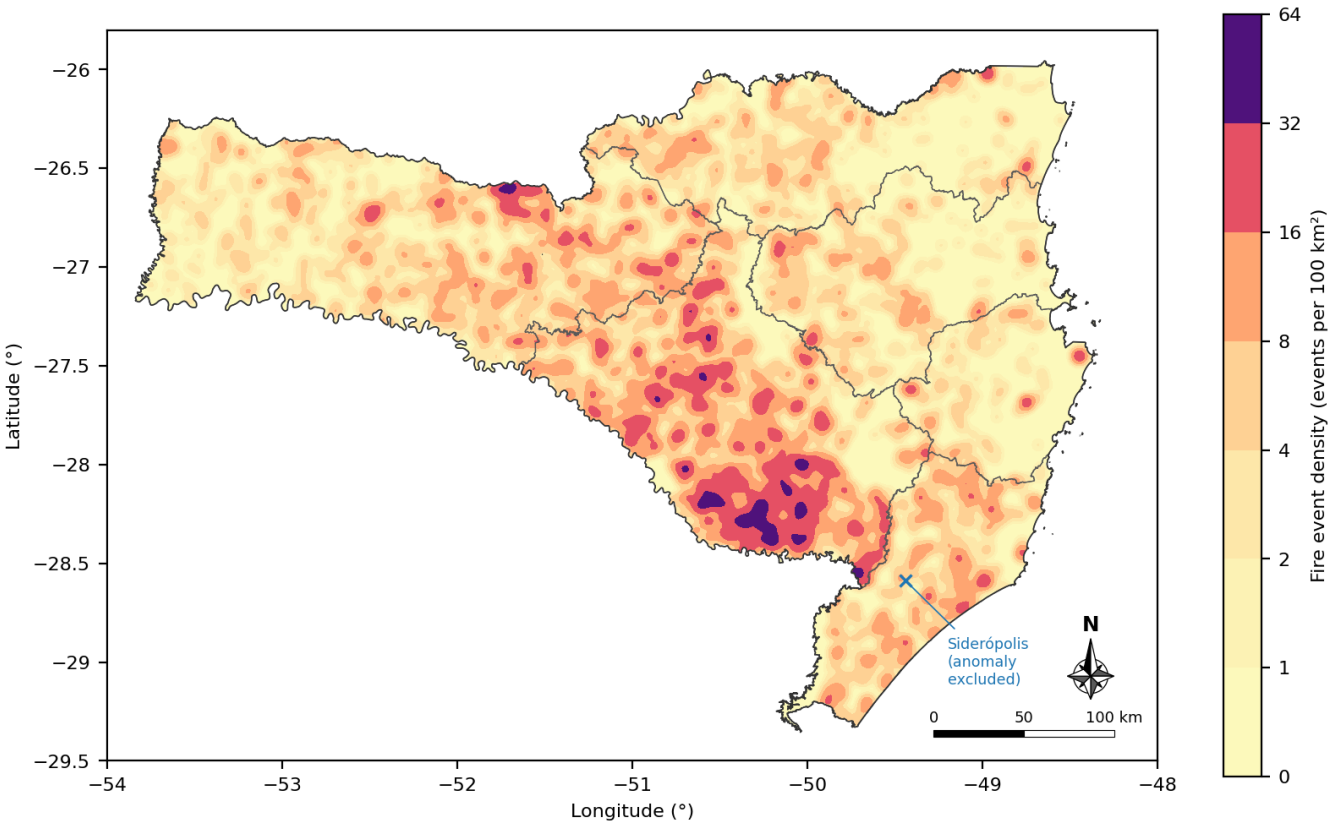


Figure 3. (a) Years with fire, 2012–2025, in each 2 km cell; (b) share of cells by dominant land cover. Script `python/analysis/fig_v71.py`; data from `a13_recurrence_landcover.py`.



Kernel density of the 2025 fire events (Supporting Information). Script `R/analysis/kde_events.R`.

CHECK YOURSELF

Which script draws Figure 3?

`python/analysis/fig_v71.py`.

8 From a published number to the code

Chapter 8 / 14 · reading time: 6 min · reviewers and co-authors

IN 30 SECONDS

Every value of the ES&T manuscript has a script, an output file and a log line. The table below lists the main ones; `docs/results_map.md` lists the numbers of the earlier manuscript and of the report.

After this chapter you will be able to:

- [ ] locate the source of a number in under a minute

Published value	Script (python/analysis/)	Output (results/tables/)
3,549–8,296 events per year (2012–2025)	`a02_series_trends.py`	`series_state_all_products.csv`
3.55 Mt CO_2 in 2025 (Table 2)	`a02_series_trends.py`	`table2_state_emissions_2012_2025.csv`
CO_2 max/min between years 2.88	`a15_v7_review.py`	`log a15_v7_review.txt`
Spearman ρ = −0.68 with precipitation	`a06_climate_drivers.py`	`climate_correlations.csv`
53.6% of grassland cells burned ≥ 7 of 14 years	`a13_recurrence_landcover.py`	`recurrence_by_landcover.csv`
VIIRS: CO_2 +39.6% to +56.2%	`a02_series_trends.py`	`viirs_effect.csv`
FINN v1.5 / v2.5 MODIS = 2.82 (2020)	`a17_v72_review.py`	`finn15_v25_2020_decomposition.csv`
Continuations 27.0% of CO_2 (2020)	`a17_v72_review.py`	`finn15_2020_continuation_rule.csv`
Tropical forest class: × 2.42 to 2.74	`a19_v72_followup.py`	`forest_class_scenario_annual.csv`
FINN area / MapBiomas Fire: median 17.2	`a05_mapbiomas_comparison.py`	`mapbiomas_state.csv`
FINN area / MCD64A1: median 34.4	`a12_gee_review_outputs.py`	`mcd64a1_comparison_summary.csv`
Earlier inventory = all FINN v1.5 records × 1.002	`a20_finn15_correction.py`	`finn15_lcqar_2020_correction.csv`
BRAIN 2019 = records × 1.010, originals × 1.390	`a20_finn15_correction.py`	`finn15_2019_correction.csv`
Continuations 27.0–32.4%, 2002–2020; Teixeira = all records	`a20_finn15_correction.py`	`finn15_2002_2020_correction.csv`
Figure 2	`figures_est_en.py`	`Fig02_annual_series_4p_en.png`
Figures 3 and 4	`fig_v71.py`	`Fig08_…, Fig09_…_nn_en.png`
TOC graphic	`figures_est_v72.py`	`TOC_graphic_est_v72.png`

DO IT: EXAMPLE, THE 3.55 MT OF CO_2 IN 2025

```
python -c "import pandas as pd; t = pd.read_csv('results/tables/
table2_state_emissions_2012_2025.csv'); print(t.loc[t.year == 2025, 'CO2'])"
# 13    3.554771e+06   (tonnes)
```

KEY IDEA

The tests in `tests/` assert these values. If a change in the code alters a published number, a test fails and names it.

CHECK YOURSELF

Where is the CO_2 of 2025?

`results/tables/table2_state_emissions_2012_2025.csv`, written by a02.

9 The statistics, explained

Chapter 9 / 14 · reading time: 12 min · scientists and students

IN 30 SECONDS

Rank-based tests (Mann–Kendall, Sen, Spearman) for short, skewed series; Holm correction for multiple tests; spatial models when neighbouring municipalities are alike; bootstrap by spatial blocks for recurrence; kernel density with several bandwidths.

After this chapter you will be able to:

- [ ] say what each test answers
- [ ] read a p value with care

Method	Question it answers	Where
Mann–Kendall test and Sen slope (Mann, 1945; Kendall, 1975; Sen, 1968)	Is there a monotonic trend over the years, and how steep?	`a02, a15`
Hamed–Rao correction (Hamed and Rao, 1998)	Does autocorrelation between years change the trend test?	`a02`
Spearman correlation with Holm correction (Holm, 1979)	Do events follow fire-season climate, testing several variables at once?	`a06`
Moran's I and spatial error model (Moran, 1950; Arraiz et al., 2010)	Is land cover associated with event density once neighbours are accounted for?	`a07, a13`
Selection ratio with spatial block bootstrap	Does fire fall on a land-cover class more than its area would suggest?	`a13, a14`
Kernel density (Diggle, 1985; Cronie and van Lieshout, 2018; Baddeley et al., 2015)	Where are events concentrated in 2025?	`kde_events.R`

KEY IDEA

With 14 years, a trend test has little power: “no significant trend” means the data cannot distinguish a trend from year-to-year noise, and the article reports it in these terms.

WATCH OUT

Ratios of two counts that share a term can correlate by construction (Kronmal, 1993). For this reason the VIIRS increment is analysed as a difference and against independent measures (`a15`).

DEEP DIVE (OPTIONAL)

Kernel density bandwidth: the main map uses σ = 2.8 km; the Cronie–van Lieshout criterion gives 8.8 km and Scott's rule 21.7 km. The concentration in the Serrana holds across these bandwidths (`kde_2025_bandwidths.csv`, `kde_2025_sensitivity_mesoregion.csv`).

CHECK YOURSELF

Why use Spearman instead of Pearson?

Fire counts are skewed and the sample is short; ranks are robust to extreme years.

10 FINN v1.5, v2.5 and the other inventories

Chapter 10 / 14 · reading time: 10 min · inventory makers and modellers

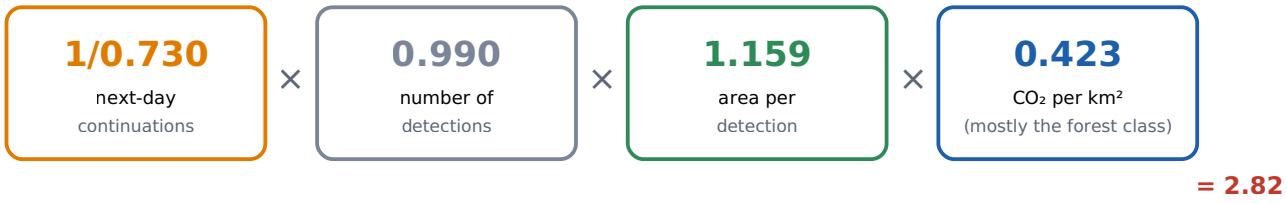
IN 30 SECONDS

For the same 2020 detections, FINN v1.5 gives 2.82 times the FINN v2.5 CO_2. The earlier state inventory, BRAIN and Teixeira et al. (2025) used FINN v1.5 with its next-day continuations; removing them lowers their totals by 27.0% to 32.4% per year.

After this chapter you will be able to:

- [ ] decompose the difference between versions
- [ ] compare an external inventory fairly

FINN v1.5 / FINN v2.5 MODIS, CO_2 of 2020 in Santa Catarina



The four factors multiply to the total ratio: the continuations (1/0.730), almost the same number of detections (0.990), 16% more area per detection (1.159) and 58% less CO_2 per km² (0.423), mostly because forest detections moved from the tropical to the temperate class. A shift-share split puts 75% of the change within vegetation classes and 25% in the partition between them (`a17_v72_review.py`).

Study (FINN v1.5)	What it equals	Without continuations
LCQAR (2022), SC 2020	all 4,609 records × 1.002 (its 4,625 “focos” are records)	1.37 × the 2,716 original detections
BRAIN 2019 (Hoinaski et al., 2024)	all records × 1.010	× 1.390
BRAIN 2020	31% of its CO on cell-days without an original detection	continuations included
Teixeira et al. (2025), 2002–2017	sum of all records in 15 of 16 years (2016: 0.36% below)	27.0–32.4% of yearly CO_2 are continuations

DEEP DIVE (OPTIONAL): REBUILDING THE FINN V1.5 RECORDS

The FINN v1.5 records of 2002, 2016 and 2017 come from the NCAR annual files. `tools/download_finn15_years.R` downloads and checks them; `tools/finn15_raw_to_grid.py` cuts each file to the State's box and converts it to tonnes; `tools/finn15_records.py` assigns the records to municipalities. On the 2015 file, this route reproduced all 3,537 rows of the earlier processing.

```
Rscript tools/download_finn15_years.R
python tools/finn15_raw_to_grid.py FINNv1.5_2016.GEOSCHEM.tar.gz 2016 data/raw_finn/finn15
python tools/finn15_records.py <authors_folder> data/raw_finn/finn15
```

⚠ WATCH OUT

Compare FINN versions in the text, with their configurations stated. Plotting FINN v1.5 and v2.5 totals on the same axis invites a direct comparison that the different conventions do not support.

CHECK YOURSELF

What share of FINN v1.5 CO_2 comes from assumed next-day fires?

27.0% to 32.4% per year in Santa Catarina, 2002–2020 (median 28.7%).

11 Route A: the inventory of your region

Chapter 11 / 14 · reading time: 8 min · users in other regions

IN 30 SECONDS

Put your boundaries in `data/inputs/`, edit `config/config.yml`, run steps 00 to 05. Start with one year: the global FINN files are large.

After this chapter you will be able to:

- [ ] configure a new region
- [ ] run route A step by step

DO IT

1. Copy the region's shapefile to `data/inputs/`.
2. In `config/config.yml`, set `region.boundaries_shp`, `region.region_field` and `region.name`, and choose years, version and sensor in `selection`.
3. Run:

```
make all-py            # or ./run_all.sh python, or run_all.bat python
python python/00_download_finn.py
python python/01_convert_units.py
python python/02_clip_region.py
python python/04_aggregate_region.py
python python/05_descriptive_analysis.py
```

Open `data/output/aggregated_emissions_all.shp` in QGIS or read the CSV of the same name.

CHECK

`tests/test_route_a.py` replicates the 2,341 records of 1 January to 28 May 2025 in Santa Catarina from the NCAR files, so route A is tested against route B.

WATCH OUT

Outside 23.5° S to 23.5° N, FINN v2.5 classes forests as temperate (chapter 4). Report this choice, and the detection source, whenever you publish totals for another region.

CHECK YOURSELF

Which file holds every parameter?

`config/config.yml`.

12 Google Colab and Earth Engine

Chapter 12 / 14 · reading time: 6 min · users without a local Python

IN 30 SECONDS

Route B runs in Colab as on any computer. The external datasets (MapBiomas, MCD64A1, ERA5-Land) come from the Earth Engine notebooks in `gee/`, which need a registered Google Cloud project.

After this chapter you will be able to:

- [ ] run route B in Colab
- [ ] run an Earth Engine notebook

DO IT

1. Register a Google Cloud project for Earth Engine at https://console.cloud.google.com/earth-engine (noncommercial use needs no billing) and note its ID.
2. In Colab, clone the repository into `/content/FINN_SC` and run route B with `--no-r`.
3. In each notebook of `gee/`, set `PROJECT = "<your project ID>"` in cell 2 and run all; cell 3 calls `ee.Authenticate()` and `ee.Initialize(project=PROJECT)`.

```
!git clone https://github.com/jessicajcss/FINN_SC.git /content/FINN_SC
%cd /content/FINN_SC
!pip install -q -r requirements.txt
!python python/run_analysis.py --no-r
!python -m pytest -q tests
```

KEY IDEA

The notebooks ending in `_run_2026-10-21.ipynb` keep the outputs of the runs used in the article, so you can compare yours cell by cell.

CHECK YOURSELF

Where do you set the Earth Engine project?

In cell 2 of each notebook, variable `PROJECT`.

13 When something goes wrong

Chapter 13 / 14 · reading time: 4 min · everyone

IN 30 SECONDS

Most problems are the environment, the working folder or a download address. The table gives the fix.

Symptom	Fix
`ModuleNotFoundError`	Activate the environment (`conda activate finn`) or install `requirements.txt` .
“file not found” for `results/` or `data/`	Run from the repository root.
A test fails after you changed code	Read the test name: it says which published number changed.
`verify_manifest.py` reports DIFFERENT	Install the pinned versions of `requirements-lock.txt` and run again; the script shows the largest differences.
FINN download returns 404	Check the address in `R/utils_finn.R` ; FINN v1.5 files are `.tar.gz` for every year.
Earth Engine asks for a project	Register one (chapter 12) and set `PROJECT` .
R step is slow	Use `--no-r` ; the annual kernel sensitivity takes about 40 minutes.

14 Glossary, references and final checklist

Chapter 14 / 14 · reading time: 5 min · everyone

IN 30 SECONDS

Terms in one line each, the sources cited in this tutorial and a six-item checklist before you report results.

Glossary

Active fire	heat detected by a satellite pixel at the time of the pass
BRAIN	Brazilian Atmospheric Inventories, a database of emissions for air-quality modelling (Hoinaski et al., 2024)
Continuation (next-day)	the half-size copy of a FINN v1.5 record added on the following day
Detection	one satellite pixel on one day (day + FIREID)
Fire event	a FINN daily burned-area polygon (day + POLYID)
GENVEG	the generic vegetation type of a FINN record
Mesoregion	IBGE grouping of municipalities; Santa Catarina has six
NRT	near-real-time files, produced daily
Record	one row of a FINN file
Route A / Route B	replication from NCAR files / reproduction from the archived records

CHECK: BEFORE YOU REPORT

- [ ] Environment installed and activated
- [ ] `run_analysis.py` finished without error
- [ ] Tests: 96 passed
- [ ] Manifest: 107 files, OK 107
- [ ] FINN data cited (models and dataset DOI)
- [ ] Configuration of FINN (version, sensor, forest class) stated with any total

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